The research program of the Center for Economic Studies (CES) produces a wide range of theoretical and empirical economic analyses that serve to improve the statistical programs of the U.S. Bureau of the Census. Many of these analyses take the form of CES research papers. The papers are intended to make the results of CES research available to economists and other interested parties in order to encourage discussion and obtain suggestions for revision before publication. The papers are unofficial and have not undergone the review accorded official Census Bureau publications. The opinions and conclusions expressed in the papers are those of the authors and do not necessarily represent those of the U.S. Bureau of the Census. Republication in whole or part must be cleared with the authors.

THE ROLE OF FINANCIAL CONGLOMERATES IN INDUSTRY FORMATION:

EVIDENCE FROM EARLY MODERN JAPAN

by

John P. Tang *
U.S. Bureau of the Census

CES 07-32 December, 2007

All papers are screened to ensure that they do not disclose confidential information. Persons who wish to obtain a copy of the paper, submit comments about the paper, or obtain general information about the series should contact Sang V. Nguyen, Editor, Discussion Papers, Center for Economic Studies, Bureau of the Census, 4600 Silver Hill Road, 2K132F, Washington, DC 20233, (301-763-1882) or INTERNET address sang.v.nguyen@census.gov.

Abstract

Large conglomerates known as zaibatsu have long been credited with leading Japanese industrialization during the Meiji Period (1868-1912). I develop a new dataset collected from corporate genealogies and estimate the likelihood of first entry with discrete choice econometric methods. I find zaibatsu are indeed more likely to pioneer new industries relative to independent firms. This may be due to their ability to finance investments internally, autonomy from shareholder interference, and lower risk aversion from being diversified. Nevertheless, zaibatsu lag independent firms in introducing innovative technologies, possibly from their preference for scale and monopolistic industries, conservative ownership, and organizational complexity.

Keywords: Meiji Period, zaibatsu, industrialization, late development, technology adoption

* Thanks to Barry Eichengreen for his encouragement and detailed comments across multiple versions of this paper. I would also like to thank Christina Romer, Bronwyn Hall, Brad DeLong, Ken Train, Rich Gilbert, Catherine Wolfram, Ted Miguel, Kenji Kushida, Rui Esteves, Chris Blattman, members of the All-UC Group in Economic History, and participants in the Berkeley development economics and industrial organization workshops for helpful comments and support. The Center for Japanese Studies and the Institute for Business and Economic Research at University of California, Berkeley provided funding for this research. All errors are mine. This paper has been screened to ensure that no confidential data are revealed. The conclusions expressed in this paper are those of the author and do not necessarily reflect the views of the U.S. Census Bureau.

The Role of Financial Conglomerates in Industry Formation: Evidence from Early Modern Japan

Abstract

Large conglomerates known as *zaibatsu* have long been credited with leading Japanese industrialization during the Meiji Period (1868-1912). I develop a new dataset collected from corporate genealogies and estimate the likelihood of first entry with discrete choice econometric methods. I find *zaibatsu* are indeed more likely to pioneer new industries relative to independent firms. This may be due to their ability to finance investments internally, autonomy from shareholder interference, and lower risk-aversion from being diversified. Nevertheless, *zaibatsu* lag independent firms in introducing innovative technologies, possibly from their preference for scale and monopolistic industries, conservative ownership, and organizational complexity.

Keywords: Meiji Period, zaibatsu, *industrialization, late development, technology adoption*

Introduction

The Meiji Period (1868-1912) witnessed a remarkable transition for the Japanese economy, whose rapid development propelled the pre-modern agrarian nation to industrialized status. Under the banner of "rich country, strong military," the economy trebled in size between 1880 and 1913 and the navy won unexpected victories against China and Russia.[1] Investment in roads, railways, harbors, and the telegraph system grew at 10 percent per year, and industrial output grew fivefold. Institutions developed apace with the establishment of a central bank in 1882, promulgation of a constitution along western traditions in 1889, and adoption of the gold standard in 1897.[2] By 1899, Japan no longer had to recognize extraterritorial rights for foreign nationals, and the country regained full tariff autonomy in 1911.[3]

How did Japan overcome its late start? Conventional wisdom credits the large conglomerates known as *zaibatsu* in "[providing] the impetus for the country's modern economic development."[4] According to this view, *zaibatsu*,[5] which emerged in the early part of the Meiji Period, had a number of advantages: size to give them sufficient scale to efficiently adopt foreign, capital-intensive technology; family ownership to have the flexibility to enter new sectors without interference from short-sighted shareholders; diversified holdings

[1] In 1895 and 1905, respectively.

[2] Lockwood, *Economic Development*, pp. 27-32.

[3] Britain was the first to renegotiate its "unequal" treaties with Japan in 1894, shortly followed by the US, Russia, France, and Germany.

[4] Morikawa, *Zaibatsu*, p. xvii. The government also played a pivotal role in industrialization by seeding strategic sectors and providing both the infrastructure and institutions that would be the basis of subsequent development; see Smith, *Political Change*; Rosovsky, "Japanese capital." Notwithstanding the government's high profile accomplishments, its importance is belied by the failure of many public enterprises and the withdrawal of direct investment in the early 1880s, before Japan's first wave of industrial activity. Its privatization program, on the other hand, contributed strongly to the development of *zaibatsu*; see Hirschmeier and Yui, *Development*; Morck and Nakamura, "Business groups."

[5] Various definitions exist for *zaibatsu*, including oligopolistic enterprises, multi-subsidiary organizations (similar to the German Konzerns), and groups of diverse firms. For the purpose of this discussion, I follow Morikawa's definition of *zaibatsu* being a family-owned diversified conglomerate; see Morikawa, *Zaibatsu*, p. xvii.

to share risk and internally finance its investments; employment of well-educated salaried managers; and access to natural resources like metals and coal.[6]

Belying these apparent advantages are a number of uncomfortable observations. First, *zaibatsu* had few opportunities to capitalize on inter-industry scale economies since their enterprises were mostly unrelated, thus limiting their ability to share resources and technologies.[7] Many of their interests were in commerce (for example, transport, trade), which were not subject to significant scale economies.[8] Even if they were able to capitalize on scale, it was only on the eve of World War I and the corresponding disruption of European trade that *zaibatsu* had a large market to serve.[9] As for their leadership in developing new industries, at a superficial level *zaibatsu* achievements are modest: the data described below indicate that of the 143 new industries[10] in the Meiji Period, only 16 were started by *zaibatsu*.[11]

Notwithstanding these problems, the visibility of *zaibatsu* has generated a substantial body of research. Numerous scholars have asserted that *zaibatsu* led the introduction and use of foreign technology in Japan's early period of industrialization, although these claims are supported mainly by anecdote, case studies, or comparisons of international development.[12] The few papers that use quantitative data to compare *zaibatsu* behavior to other firms are limited to financial records dating from the interwar years, after the Meiji Period. The paucity of analysis, stemming from a lack of data, leaves as a mystery much of what helped *zaibatsu* and Japan to succeed.

This paper attempts to fill in some of these gaps, in particular on the issue of technological leadership. I test the hypothesis that *zaibatsu* were more likely to be pioneers in new industries compared to their independently established rivals, controlling for other characteristics such as ownership and location. My main finding is that conglomerate affiliation does increase the likelihood that a firm will be a first entrant in a new industry. An interpretation of this result is that an affiliated establishment has access to internal financing and/or lower cost capital compared to a standalone firm, which may explain the stronger correlation

[6]Ibid., pp. 12-14; Fruin, *Japanese Enterprise System*, pp. 3-5. These advantages are magnified in a Gerschenkronian late developing economy; in other words, one with weak institutions, poor infrastructure, and immature capital markets; see Gerschenkron, *Economic Backwardness*. So powerful were these industrial cliques that they dominated the economy until the end of World War II. In the post-war era, after a brief interregnum when the American occupation authorities disbanded them, they were reincarnated to help Japan rapidly re-industrialize and attain its "miraculous" recovery. See Dodwell, *Industrial Groupings*; Morikawa, *Zaibatsu*.

[7]Fruin, *Japanese Enterprise System*. This is true for the Meiji Period; in the 1900s *zaibatsu* engaged in more manufacturing activities, which allowed for scale economies.

[8]While commerce did allow for economies of scope, the attributes of size and wealth are less meaningful. Scope economies differ from scale economies in their reliance on the savings from fixed costs (for example, shared facilities, distribution channels) rather than variable costs (for example, shared inputs, learning curves). Another way of distinguishing the two is that scope economies typically involve production of multiple, unrelated goods while scale economies are usually from increased production of the same (or similar) goods. Ibid.

[9]Morikawa, *Zaibatsu*.

[10]Industries are measured at the four-digit industry classification level; further discussion of the data is in later sections. See the appendix for a complete list of new industries established in the Meiji Period.

[11]That said, it can be argued that the absolute number of first-entry firms does not adequately capture economic impact due to differences among industries (such as number of entrants). For example, later entrants in an industry can still lead in scale of operations. As I discuss later, a *zaibatsu* affiliate typically was much larger and produced more than a single independent firm, although missing data prevent a comprehensive comparison (and thus motivates this research).

[12]Fruin, *Japanese Enterprise System*; Morikawa, *Zaibatsu*.

between *zaibatsu* and capital-intensive industries.

I also find that regardless of affiliation, a publicly-listed establishment (for example, joint stock corporation) is less likely to be an industry pioneer than a privately-held establishment. This corresponds with the late development theory that financial system immaturity may impede industrialization absent public intervention or exceptional institutions to lead development.[13] In this view, *zaibatsu* were well-equipped for leadership, having access both to their wealthy owners' assets and to internal financing from existing holdings.

While the above findings are consistent with conventional wisdom about the role of *zaibatsu*, it also appears that *zaibatsu* lag other firms in the introduction of highly innovative technology (in other words, technology dissimilar from any existing in the economy), with the effect becoming more pronounced over time.[14] This may be due to growing conservatism among *zaibatsu* owners, with firm founders seeking to protect the family patrimony as well as later generations inheriting wealth but not entrepreneurial spirit. Increased organizational complexity arising from excessive diversification may have also weakened the desire for continued innovation (and offset any benefits from further diversification).

At the heart of this analysis is the assumption that the first appearance of an industry using new technology is a reasonable approximation of when that technology was introduced to Japan.[15] As a late developing country, Japan was able to borrow existing technologies without needing to develop them itself, and thus the first appearance of an industry using new technology proxies the technology's introduction to the country. My analysis eschews the need for financial records, few of which existed before the twentieth century, developing instead a new dataset consisting of firm establishment dates from the Meiji Period that I gathered from corporate genealogies. By studying the order of technology introduction via industry establishment, I can determine whether a firm's affiliation (*zaibatsu* or not) had an impact on the likelihood of it being the industry pioneer. This in turn should be indicative of the role of conglomerate membership in industrialization.

In addition to the new dataset, this study of *zaibatsu* and Japanese industrialization improves on the existing literature in several ways. First, I use a broad sample of *zaibatsu*, which is logical given that discussions about trends in Japanese industrial development generally refer to them as a group.[16] I then examine *zaibatsu* influence across multiple industries classes rather than in a specific industry to assess the economy-wide impact of these corporate groups.[17] Finally, I focus on the Meiji Period, when Japan first

[13] See Teranishi, "Were banks?" for a discussion of prewar Japanese banking.

[14] This behavioral change is much more apparent in the decades following the Meiji Period; see Frankl, "Analysis."

[15] Nevertheless, outside the late development context, there is an important difference between the development of new technology and its application, since developers may not have the resources to bring the technology into production (for example, modern-day research laboratories versus large pharmaceutical companies). This is not to say that the first firm to introduce new technology will necessarily succeed, as other firms may prefer to observe market reception before committing their resources or to learn from the experience of the first firm. Some of these issues will be explored in more detail in the later sections.

[16] *Zaibatsu* used for the analysis include Mitsui, Mitsubishi, Sumitomo, Yasuda, Furukawa, and Ōkura. These six are the biggest and oldest *zaibatsu* established in the pre- and early Meiji Period (with the first four referred to generally as 'The Big Four'), and their grouping together for analysis is consistent with Japanese practice in differentiating older from newer *zaibatsu* that emerged in the 1900s; see Frankl, "Analysis."

[17] Industry class refers to the classification of industries at varying degrees of specificity; in other words, two- through four-

began to industrialize, whereas other authors may have been forced to draw inappropriate conclusions about this crucial period from later periods due to their reliance on financial records.[18]

The Significance of *Zaibatsu*

Research on *zaibatsu* has developed around three major themes: their role in Japanese industrialization, their relationship with the government, and their performance relative to independent firms.[19] In their studies of Japan before the Pacific war, Hidemasa Morikawa and Mark Fruin argue that *zaibatsu* took the lead in introducing foreign technology by employing of students who either studied abroad or graduated from the newly established universities teaching occidental sciences; cultivating contracts with foreign manufacturers to import capital equipment and skills; and reengineering western technology to suit local resources and market conditions.[20] *Zaibatsu* achievements in Japan include the first modern steel ship, the first insurance company, and the first multidivisional (M-form) corporation. Table 1 provides a list of industries entered and pioneered by *zaibatsu*.

Keiichiro Nakagawa suggests that government patronage accounted for the emergence of manufacturing firms by providing both the social and physical infrastructure needed by entrepreneurs and the initial investment in western technology and equipment.[21] The Meiji government, for its part, subsidized foreign education and employed foreign experts to work and teach, supplying administrators and engineers to *zaibatsu*.[22] Moreover, the Sino-Japanese (1894-1895) and the Russo-Japanese (1904-1905) wars enabled well-connected businessmen to procure supply contracts in shipping, construction, armaments, and mining.[23]

Three recent papers compare the performance of *zaibatsu* and independent firms in the 1900s. Jennifer Frankl, using interwar (1915-37) financial records for 100 firms, analyzes the effect of *zaibatsu* affiliation on equity returns and risk profiles. She finds that Meiji-era *zaibatsu* had less stability in their returns on equity than both independent firms and the newer *zaibatsu* of the Taisho (1912-26) and Showa (1926-89) Periods.

digit industry classification codes. For example, a two-digit code of 05 refers to Metal Mining; a three-digit code of 053 refers to Iron Ore Mining; and a four-digit code of 0534 refers to Chromium (a type of iron ore) Mining. More discussion about the data and its coding is in the body text.

[18]This is important not only to better understand the genesis of modern Japanese industries, but also to circumvent the distortions associated with the global depression in the 1920s and militarization in the 1930s. Ohkawa and Rosovsky characterize Japanese modernization in terms of "recurrent waves" or "long swing expansions;" in other words, Kuznets cycles of expansion and retrenchment; see Ohkawa and Rosovsky, *Japanese Economic Growth*. According to them, the first wave of Japanese industrialization began in 1901 and ended in 1917. The Commercial Code of 1893 established the modern Japanese corporate system based on ownership, for example, unlimited liability, joint stock firm (Loenholm 1906).

[19]A comprehensive survey of Japanese business history that provides a context for *zaibatsu* development can be found in the 15 volume series of proceedings from the Fuji Conferences published by the University of Tokyo.

[20]Morikawa, *Zaibatsu*, pp. xvii-xxiv; Fruin, *Japanese Enterprise System*, pp. 3-5.

[21]See Nakagawa, "Business strategy," pp. 3-12. This view is controversial, with authors like Morikawa arguing that there are a number of *zaibatsu* that nearly collapsed due to the vagaries of political patronage as well as arose without recourse to political mercantilism.

[22]Morikawa, *Zaibatsu*; Jones, *Live Machines*. The French engineer Paul Brunat, who was responsible for managing the government-built Tomioka Silk Reeling factory, was paid $600 monthly in current prices, equivalent to that of government ministers, and foreign silk reelers were paid $80 per month, 50 times the wage of domestic reelers; see Kiyokawa, "Transplantation."

[23]Lockwood, *Economic Development*; Yamamura, "Success?"

In contrast, Tetsuo Okazaki, using financial data from 1922 to 1936 for 135 firms, finds that those affiliated with *zaibatsu* outperformed independent firms and attributes this to the holding company organization of the former.[24] These results are supported by Hideaki Miyajima, Yusuke Omi, and Nao Saito, who conclude that concentrated ownership of a firm corresponded with better returns and that *zaibatsu* had less volatile returns.[25] Nevertheless, because of the limitations of their data, all three papers are constrained to analyze the twentieth century, when Japanese industrialization was already underway. They also focus on the holding company characteristic of *zaibatsu* without considering structural features of the industries and make only passing reference to technological introduction and leadership.

Example: Mitsubishi and Japan's Maritime Industries

It may come as a surprise that an island nation like Japan would not develop modern shipping and shipbuilding industries until the late 1800s. In fact, Japan had both, although its shipping industry was confined to domestic waters and its shipyards to construction of wooden ships no larger than 75 feet in length or 150 tons in weight.[26] Moreover, the development of these two industries and that of the Mitsubishi *zaibatsu* are closely intertwined.

The modern shipping industry began in 1870 with the establishment of Tsukumo Shōkai, later renamed Mitsubishi Shipping Company. This company was the first of many in the Mitsubishi *zaibatsu* led by founder Iwasaki Yatarō, and initially served to intermediate between foreign and native merchants as well as to procure foreign-built ships. Soon after its 1875 inaugural overseas route was between Yokohama and Shanghai, the company expanded rapidly along the coast, then to Mumbai in 1894, and to London, San Francisco, and Australia in 1896.[27] While these early journeys were mainly for postal deliveries, the diversification of Mitsubishi meant that business increasingly was in the goods trade. However, strong competition with British and American shipping prevented Japan from developing large-scale international operations until World War I, which substantially decreased international shipping capacity and left open market opportunities for Mitsubishi and other Japanese shippers.[28]

The lack of technology and facilities to build modern steel ships meant that the shipyard industry developed after the shipping industry, since the latter could and did import foreign-built ships for their business in its early years. The advent of a modern shipyard industry came about in 1895, when the first steel steamship *Suma Maru*, at 1,522 tons, was built.[29] This accomplishment

[24]This is due to the efficiency of internal monitoring of firms by the holding company, as opposed to shareholder monitoring of publicly-listed firms; see Okazaki, "Role."

[25]Miyajima et al., "Corporate governance." Of the approximately 600 firms in Miyajima's sample, 50 have records between 1900 and 1912 (in other words, the late Meiji Period).

[26]These limits were due to a series of protectionist edicts at the beginning of the Tokugawa Period (1603-1868), where the ruling government under Shogun Tokugawa Iemitsu banned large ship construction in order to isolate the country from foreign influence and trade.

[27]An earlier, non-commercial international voyage was in 1874, when the Meiji government commissioned Mitsubishi to transport military troops to Taiwan for a punitive attack on that island's aborigines.

[28]Mitsubishi monopolized the overseas shipping industry until 1891, when Osaka Shipping Company extended its domestic postal shipping service to Korea.

[29]However, the first modern ship of notable size was built in 1898, called the *Hitachi Maru* at 6,172 tons.

was also at the hands of the Mitsubishi *zaibatsu*, which owned the Nagasaki Shipyard that built the vessel.[30] Twenty-five years later, Japan had become the third largest shipbuilding nation, following the United Kingdom and the United States, with a fleet of 1,940 ships totaling almost three million gross tons in weight.[31] One constraint to the early development of the shipyard industry was a lack of domestic raw materials for construction. Fortunately, resources like iron ore and coal were available in Southeast Asia and northern China. The efficiency and advancement of this industry were such that by WWII, construction costs were a third less than its nearest rivals in Britain and Germany and half the cost of an equivalent American ship, savings which were driven primarily by low labor costs.[32]

Entry Timing and Innovation

Industry pioneers are able to establish new markets, garner market power and monopoly profits, and set industry standards. Additionally, firms in capital-intensive or high minimum-efficient-scale industries benefit from lower average costs as they increase production, which is easier to do the fewer the number of competitors. First movers may gain a head start in the learning process as they acquire experience, which can also lower production costs, and develop linkages with suppliers and distributors to cement their market leadership. Understanding the importance of firm characteristics like conglomerate membership may thus help to clarify the roles of size, ownership, and organization in late development and economic catch-up.

In a widely-cited survey, Marvin Lieberman and David Montgomery discuss three factors favoring market pioneers: technological leadership, resource preemption, and switching costs.[33] Technological leadership can arise either through a head start in learning-by-doing or through research barriers such as patents.[34] A first mover can also dominate a market by being the first to acquire scarce resources, be they physical, financial, human, or even geographic.[35] Finally, early entrants are likely to sustain market leadership if consumers face high transaction costs or have incompatible sunk investments when switching producers.[36]

At the same time, early entrants face market and technological uncertainties and competition from followers who can free-ride on incumbent investments. In a study of the American animation industry, Alan

[30]The government first built this shipyard in 1871, but sold it to Mitsubishi in 1887.

[31]Even so, engines and turbines to power these ships continued to be imported until after WWI.

[32]Morikawa, *Zaibatsu*; Travis, *Shipping Industry*.

[33]Lieberman and Montgomery, "First mover advantage."

[34]See Lilien and Yoon, "Timing," on the importance of research investment for industry pioneers.

[35]For more recent studies on resource constraints, see Robinson et al., "First-mover advantages," on high initial costs and Fuentelsaz et al., "Followers' entry timing," on geography.

[36]More recent work has underscored the effectiveness of entry barriers, although there is substantial variation depending on the particular barrier; see Han et al., "Entry barriers." Consistent with earlier research, they find that incumbent firms can deter competitive entry most effectively through the use of proprietary assets and production cost advantages; see also Mitchell, "Whether." Others find that sectors with more first entry advantages tend to be developed earlier; see Schoenecker and Cooper, "Role." This occurs partly from widespread recognition of potential profits, encouraging a race for first entry. They also report earlier entry for larger firms, those with active marketing, and those with greater access to technology. There appears to be no timing advantage in having greater financial resources or diversity of operations. Incumbent firms are also more likely to expand into a new, related sector if they perceive potential competition, and not to diversify (for fear of cannibalizing existing sales) absent that threat. One significant limitation to this study, however, is the exclusion of startup firms that are established for a new sector since the authors wanted to compare existing features of potential entrants.

Bryman finds that follower firms outperformed earlier movers due to the failure of the latter to adapt to changing tastes.[37] Jamal Shamsie, Corey Phelps, and Jerome Kuperman find that latecomer firms are more successful if they are large and draw on pre-existing resources, regardless of market conditions like industry competitiveness.[38]

An established firm's ability to reallocate resources away from unsuccessful new ventures can attenuate the risk of failure. Using a model of entrepreneurship, Denis Gromb and David Scharfstein suggest that skilled workers take into account the strength of the external labor market when choosing between entrepreneurship and working in an established firm.[39] If entrepreneurial activity is high and the pool of human capital is large, skilled workers do not need the safety net provided by working in an existing firm. However, if the entrepreneurial labor market is weak, then the high cost of a failed venture may deter startup activity.[40]

The decision to invest in a risky venture (for both entrepreneurs and established firms) also depends on industry characteristics. Using U.S. manufacturing startup data from 1976 to 1986, David Audretsch provides evidence that entrepreneurs are more likely to start a company in industries that have greater knowledge asymmetries or exploit new technologies.[41] This is because entrepreneurs in these new industries are better able to appropriate the value of their innovation than entrepreneurs working within an existing firm.

These theories do not yield clear predictions of whether *zaibatsu* were more likely to be industry pioneers or laggards. Clearly, *zaibatsu* had the financial means and autonomy to invest in new sectors, and failures in the labor and capital markets enhance the advantages of large, established companies. To clarify the extent that differences between *zaibatsu* and independent firms mattered for economic development, I use a new dataset of firm establishment to test the hypothesis that *zaibatsu* were more likely to lead entry into innovative industries.

Data and Methodology

The primary data source is the *Shuyo Kigyo no Keifuzu*, a compilation of corporate genealogies edited by the business historians Shintaro Yagura and Yoshiro Ikushima.[42] Additional sources of data include firm financial reports from the *Eigyo Hokokusho Shusei* collection,[43] which give balance sheet data for publicly-listed firms, including capitalization value, revenues, profits, assets, and liabilities; the 1984 edition of the *Standard Industrial Classification of Japan*,[44] which provides industry codes for each observed establishment; the US *Census of Manufactures* from 1905,[45] which provide labor and capital expenditures to measure differences

[37] Bryman, "Animating."
[38] Shamsie et al., "Better late."
[39] Gromb and Scharfstein, "Entrepreneurship."
[40] Similar reasoning applies to the redeployment of financial capital; see Gertner et al., "Internal versus external markets."
[41] Audretsch, "Asymmetric information."
[42] Yagura and Ikushima, *Shuyo*.
[43] Yushodo, *Eigyo*. Given the scarcity of reports from the Meiji Period, most of the firms in the collection postdate those in the current dataset.
[44] Statistics Bureau of Japan, *Standard Industrial Classification*.
[45] US Census Bureau, *Census*.

in industry factor intensity; and various Japanese industry indices and firm case studies.[46]

A common problem to many historical studies of economic development is a lack of reliable and comprehensive data. Few developing countries have the capacity to create and maintain detailed information on industries and firms, and Japan in the nineteenth century was no exception.[47] Nevertheless, an alternative can be found in corporate genealogies, which can trace a firm's origins and provide basic information like a date of establishment, ownership and source of setup finance, industry classification, and geographic location. Despite being less quantitative than financial measures, these data are sufficiently detailed to indicate relationships between specific sectors while controlling for the abovementioned characteristics. More importantly, they oftentimes represent the oldest reliable evidence of firm activity across the industrial spectrum. Figure 1 shows an example of a firm genealogy and the information each observation typically provides.[48]

The *Shuyo* compilation includes genealogies for 1,089 firms that were listed on the Tokyo Stock Exchange as of September 1984, and cumulatively contain over 14,000 unique establishments dating back to the early nineteenth century or prior.[49] These genealogies are a representative sample in that they not only include all ancestor firms of those surviving to 1984, but also acquisitions and asset transfers from firms outside direct lineages.[50] The inclusion of these firms, which otherwise would not appear in any other historical record, to an extent minimizes the issue of firm survivor bias. In addition, it can be argued that manufacturing, be it in textiles or machinery, is the primary focus of industrialization. Unlike services, which may require little capital investment or equipment, manufacturing typically leaves behind assets that may be purchased by other manufacturers, thus increasing the chance that the previous owner will appear in the genealogies. Figure 2 shows the number of startup establishments over the Meiji Period, between 1868 and 1912.

Industry identification is provided directly through the company name or descriptor annotated to the firm, with which I retroactively apply the codes given in the 1984 edition of the *Standard Industrial Classification for Japan* (JSIC).[51] Typically, company names in Japan are in three parts: personal/geographic name + industrial activity + industrial operation/facility (for example, Ishitsuka + Bottle Manufacturing + Factory), with the most common company names using a combination of the first two identifiers. Of course, there are some exceptions to the single-industry-per-establishment identification, excluding conglomerates. For example, Uemo Coach and Rail is classified as both in the Local Railway (JSIC three-digit code 402) and Light Passenger Vehicle Transport (JSIC3 414) industries.[52]

[46]For example, Asia Research, *Outline*; Dodwell, *Industrial Groupings*; Mitsubishi Public Affairs Committee, *Brief History*; Ohsono, *Charting*; Russell, *House*.

[47]For example, the *Long Term Economic Statistics of Japan* begins its industrial series in 1887 and are only available at the national level; see Ohkawa, *Nihon*.

[48]Not all the observations contain every piece of information, for example, some lack their establishment location. The discrepancies are apparent in the different sample sizes provided in the summary statistics and econometric results.

[49]Yagura and Ikushima, *Shuyo*. Note the difference between establishment and firm, with the latter possibly including multiple establishments (like *zaibatsu*). Due to the breakdown by industry, firms with multi-industry production are each given a unique identifier, but entered multiple times given the number of different industries.

[50]The dataset used in this paper contains only independent firm startup activity and does not indicate a firm's subsequent success or failure.

[51]The Japanese SIC system resembles the defunct American SIC system (replaced with the North American Industry Classification System), but is not a one-to-one mapping.

[52]Also, the move toward abbreviation, multiple personal names, and deletion of industrial activity has largely occurred in

Although the classification of Japanese industrial sectors did not begin until 1930 and has been revised a number of times since, retroactive classification can be rationalized based on the following reasons. First, a lack of an industrial classification system in the Meiji Period means retroactively applied codes do not alter the historical record. Second, industrial sector distinctions that were made in later years do not preclude the existence of those distinctions during the Meiji Period. Third, codes for industries that did not exist in the Meiji Period do not have to be used. Fourth, industries that existed in the past that do not appear in the 1984 system can be additively included without needing to change existing codes.

The premise of this study is that firms affiliated with *zaibatsu* and firms established independently differ in fundamental ways, with implications for the development of industries and the introduction of technology. Differences include access to natural resources (for example, coal, iron); managerial autonomy; the ability to finance investments internally; risk-sharing from diversification; the employment of highly skilled labor; and relationships with the central government. These characteristics are assumed to influence if and when firms pioneer industries. Other characteristics such as regulatory environment and market demand are taken as common to both types of firms.[53]

For my analysis, I use a discrete-choice probit regression model to estimate relative likelihoods of entry. Under the hypothesis that *zaibatsu*-affiliated firms are more likely to be first entrants in new industries, I use the entry outcome (first entry or not) as my dichotomous dependent variable. I include the following independent variables to determine the relative influence each plays in the choice to be a first entrant: conglomerate affiliation, firm ownership type, the number of industries a firm is operating in at the time of entry (in other words, diversification), industry innovativeness, industry factor intensity, and the population density of the prefecture in which the firm establishes. I also have dummy variables for major industry type and interaction terms between variables.

The key independent variable is firm affiliation, which takes the values of zero for independent establishment (in other words, startup) or one for membership in a *zaibatsu*. I include affiliates of all the major *zaibatsu* established in the first half of the Meiji Period or earlier: Mitsui, Mitsubishi, Sumitomo, Yasuda, Furukawa, and Ōkura.[54] This variable captures unobserved differences between a *zaibatsu* firm and an independent one, such as lower capital costs, internal financing, information spillovers, etc. Under different specifications of the regression model (in other words, the inclusion of different independent and control variables), I can compare observable characteristics of conglomerate membership (for example, ownership, diversification) to unobservable ones (for example, internal financing, network externalities) and determine their contributions to first entry. I hypothesize that *zaibatsu* affiliation has a significant positive correlation with first entry.

The variable for firm ownership type takes the value of one for publicly-listed (joint stock) firms or zero for privately-held firms.[55] While the analysis distinguishes only two types of ownership, there are a

the post-World War II period; see Yagura and Ikushima, *Shuyo*.

[53]Nevertheless, considering the lack of specific firm data like revenues and market share from this period, estimation of this reduced set of variables may be problematic. Other variables help to clarify what features are not captured by conglomerate affiliation and contribute explanatory power.

[54]Cumulatively, there are 58 *zaibatsu* affiliates in the dataset.

[55]There may be some confusion as to terminology: "privately-held" means firm equity that is not available to the public as

number of variations: private ownership includes individual proprietorship or partnership (unlimited and limited liability) as well as mutual associations, and public firms came in both limited and unlimited liability flavors.[56] Given that private ownership was one of the defining characteristics of *zaibatsu*, this variable may indicate the importance of investment autonomy and external monitoring.[57] In an economy with an immature financial system, there may be fewer advantages of the joint stock form to finance investment compared to internal or debt-based financing. Anecdotal evidence suggests that equity-financed firms were constrained by their need to pay dividends, resulting in firms that remained small and undercapitalized.[58] Furthermore, privately-held ownership may allow a firm to make longer-term investments since financing was neither subject to business cycle volatility nor reliant on investors unwilling to tolerate long gestations until the enterprise makes a profit.

This variable also addresses the issue of corporate monitoring, prominent in discussions of postwar Japanese conglomerates (although important in the prewar era as well). Prior to the adoption of the 1893 Commercial Code, which standardized incorporation procedures and defined fiduciary responsibilities, the limited ability of outside investors to monitor management and dominant owners may have hindered the public listing of firms (and thus created a market failure for investment).[59] This is because while incorporation occurred as early as 1868, the lack of institutions governing business practice or protection of property rights remained until the 1890s.[60] Together these observations suggest a positive correlation between private ownership and first entry.

The variable for industry diversification is the number of industries in which a firm is operating in at the time it establishes an enterprise in a new industry;[61] in the dataset, this variable ranges from 1 to 21 (in other words, for the Mitsui *zaibatsu*). By definition, a conglomerate is a firm that operates in multiple industries. In the dataset, there are also independent firms that operate in multiple industries, but this occurs usually at the time of entry (in other words, simultaneous entry into two related industries). Arguably, diversifying across industries reduces volatility in revenues and spreads industry-specific risk. On the other hand, having many different industry holdings increases administrative complexity and the potential for inter-divisional conflicts in management and strategy.[62] Nevertheless, it is reasonable to expect that a diversified firm is more likely to invest in a risky industry relative to an independent firm.

The rationale for focusing on industry innovativeness is that technologies with precedents in the market are less likely to be rejected by the public (since the first technology of its kind was already introduced

shares (in other words, unlisted), and differs from "private sector" firms, which are those not owned by the government.

[56] Yagura and Ikushima, *Shuyo*.

[57] While there may be some overlap between private ownership and conglomerate affiliation (since conglomerates were largely private), the two variables are different in that there were a number of non-zaibatsu investors who owned private firms while some *zaibatsu* held equity in publicly-listed firms.

[58] Morikawa, *Zaibatsu*; Teranishi, "Main bank system." For example, many publicly-listed firms were run for short-term profit and were incorporated for a predetermined time period, between three to ten years; see Fruin, *Japanese Enterprise System*.

[59] Loenholm, *Commercial Code*. There is an interesting literature on the effect of owner-managed firms on performance; see Denis et al (1999).

[60] Rosovsky, *Capital Formation*.

[61] At the four-digit level.

[62] There were many disputes, for example, within the Mitsui *zaibatsu* between the directors of the trading company and the bank over investment strategy in the late 1800s; see Morikawa, *Zaibatsu*.

and thus familiar) or are more transparent in their operational difficulty and expense, thus representing a lower investment risk. The innovativeness variable takes a value of one for an industry that is the first to be established out of its broader industry grouping, and zero for industries that are not. An example of this is the three-digit industry grouping "251: Glass Manufacturing," which includes specific industries at the four-digit level like "2511: Plate Glass," "2514: Glass Container," and "2515: Scientific Glassware." If there were no glass manufacturing industries prior to 1871, when the Ishitsuka Glass Container Company was founded, then the four-digit industry "2514" would be coded as innovative, and the other four-digit glass industries would be coded as not.

Furthermore, with an industry code I am able to determine the factor intensity of a given industry. Although Japanese statistics do not provide sufficient detail about historical industries, I use capital-labor expenditure ratios from contemporaneous American industries in order to gauge the effect it had on firm entry.[63] The weakness of private capital and lack of robust financial intermediaries suggest difficulties to mobilize large amounts of capital for scale-oriented and high fixed-cost industries, which may have incited *zaibatsu* activity in such sectors early in the period.

While Japan has long been known as a densely populated country, under-appreciated is the effect of topography on population distribution. A naïve calculation of population density would take a prefecture's population and divide by its surface area, but this grossly underestimates actual density. That is because less than 20 percent of the country can be cultivated and the majority of settlements are in less elevated, alluvial plains.[64] If one considers only the proportion of land in each prefecture that is at lower elevations, Japan's population density increases markedly.[65] This adjustment is important because urban areas differ from rural ones in market size, infrastructure, commercial and industrial networks, political influence, and health outcomes.[66] In particular, access to internal financing among *zaibatsu* may be less an advantage in more densely populated areas.

Based on population density, I can also distinguish between urban and rural areas, with urban being any prefecture with over 386 people per square kilometer.[67] Thus, 26 prefectures can be classified as urban at the beginning of the period in 1868.[68] This number increases to 36 by the end of the period in 1912.

I control for industry-level differences with dummy variables for four general industry categories: agriculture and construction; manufacturing; utilities, transportation and communication industries; and financial services. Separating industries by type is important due to differences in capital requirements, scale, and

[63] I create capital-labor expenditure ratios by dividing operating costs (materials and energy) by labor wages, figures from the 1905 edition of the *Census of Manufactures*; see US Census Bureau, *Census*. While the correspondence applies primarily to manufacturing, there are some non-manufacturing establishment entries that have capital-labor ratios because of availability in the Census data.

[64] Trewartha, *Japan*.

[65] Japan's Ministry of Land, Infrastructure, and Transport has four general topographic classes: mountain, hill, upland, and lowland. For the purpose of calculating population density, I use the sum of upland and lowland areas by prefecture; see Statistics Bureau of Japan, "Land."

[66] Trewartha, *Japan*.

[67] This number is calculated using the US Census' definition of urban as an area with a population density exceeding 1,000 inhabitants per square mile; see Long et al., "Population density."

[68] Population figures for prefectures are available starting in 1884, with earlier years using estimates extrapolated from those figures; see Ohkawa, *Nihon*.

other characteristics shared within industry families but not across all industries. I also cluster the standard errors in the regressions using the four-digit JSIC industry codes. This is to account for random industry-specific shocks that are shared within narrower industry groupings (for example, drought for agricultural industries).

To identify shared influences, I interact *zaibatsu* affiliation and ownership with each other and with industry innovativeness, industry type, factor intensity, and population density. In particular, *zaibatsu* affiliation and privately-held ownership, although generally identified with each other, are not identical. A number of *zaibatsu* held substantial shares in some publicly-listed firms, such as Sanyo Railway and Japan Postal Shipping Company, but did not exercise control, and other privately-held firms were not affiliated with *zaibatsu*. The interaction between the two variables may show differences in the behavior of *zaibatsu*-controlled firms versus those that simply had a *zaibatsu* connection (and possibly access to *zaibatsu* capital). I anticipate a positive correlation between privately-held firms affiliated with *zaibatsu* and first entry into new industries, consistent with earlier mentioned benefits of both conglomerate membership (for example, risk-sharing, credit access) and private ownership (investment autonomy, long-term planning).

I remove government firms from the sample on the grounds that the behavior of such firms is not obviously driven by market factors. I also remove all industries in which the government was the pioneer to focus on the behavior of private sector firms that were operating under market forces. Additionally, I include only sectors that were established in the Meiji Period since pre-Meiji industries are less likely to use technology borrowed from abroad. Similarly, to lessen possible bias from undocumented earlier establishments in traditional sectors, I have excluded all retail and miscellaneous service industries (for example, merchandising, recreational establishments), miscellaneous financial activities (for example, pawnshops, moneychangers), and traditional industries like Japanese apparel and alcohol. Furthermore, establishments in foreign countries are omitted as well. For duplicate appearances in the dataset due to changes in name or ownership, only the first appearance is included in the analysis.

Results

In the period from 1868 to 1912, 1,958 entrants could be identified by a four-digit JSIC code.[69] After imposing the restrictions mentioned above, the dataset has 1,481 entrants, of which 1,449 were independent firms and 32 were affiliated with *zaibatsu*. The sample covers 111 industries at the four-digit industry level, of which 12 have *zaibatsu* entry. Additional summary statistics are in Table 2, with the left panel consisting of establishment-level observations and the right panel of first entries into four-digit industries.[70]

There are some interesting differences between *zaibatsu* and independent firms. At the establishment level, *zaibatsu* are disproportionately represented in innovative industries (four-fifths of their total entries)

[69] Entrants include both individual firms as well as industry divisions within multi-industry companies (for example, conglomerates).

[70] While the relative numbers of *zaibatsu* to independent firms suggest an imbalance in the sample, the absolute number of firms belies substantial organizational and productive differences. Nevertheless, to avoid possible over-representation bias, I perform separate regressions on a subset of industries contested by both *zaibatsu* and independent firms. The results are reported in the section on robustness checks.

compared to their independent rivals (one-fifth). Among broad industry groups, over two-thirds of non-*zaibatsu* establishments are in the financial sector compared to one-third for *zaibatsu*. These conglomerates are also predominantly privately-held firms unlike independent establishments.

The picture changes somewhat when looking at the industry level. Notwithstanding a modest number of establishments in manufacturing compared to financial services, independent entrepreneurs prefer to lead entry in the former (over half of all their first entries). In contrast, *zaibatsu* favor leadership in service industries (for example, transportation, retail, finance), which may be surprising considering the financial advantage of *zaibatsu* and the investment requirements for capital-intensive sectors. Historically, however, this is reasonable given that all six *zaibatsu* emerged from extractive or service industries and did not pursue heavy industrial activities until after the Meiji Period.[71] Interestingly, none of the *zaibatsu* had establishments in rural prefectures, which may be due better distribution networks based in cities compared to single establishment independent firms.

Correlations

Results from pairwise correlation analysis in Table 3 are largely consistent with the summary statistics. As hypothesized, first entry is positively correlated with *zaibatsu* affiliation, industry diversification, and manufacturing.[72] It is negatively correlated with public listing on an equity market and financial services.

Besides first entry, *zaibatsu* affiliation has positive correlations with industry diversification, industry innovativeness, the primary sector, and capital-intensive industries. These results correspond with the historical development of *zaibatsu*, with two having substantial mining interests (Sumitomo, Furukawa) and two in shipping and trade (Mitsui, Mitsubishi). The negative relationships with public listing and financial services buttress the earlier claim that *zaibatsu* had access to internal financing and lower borrowing costs (despite the Yasuda *zaibatsu* being almost exclusively a financial services firm). This suggests that these conglomerates did not need to establish financial service firms to acquire cheap capital for investment during the Meiji Period.

Diversification is positively correlated with innovative industries, which makes sense given the risk-sharing benefit of multi-sector operations, and all broad industry groups except financial services. It is negatively associated with public listing, possibly for the same reasons as *zaibatsu*. Publicly-listed ownership is negatively correlated with industry innovativeness and most industries including manufacturing, while being positively correlated with financial services. An explanation may be the over-representation of *zaibatsu* among privately-held firms (which will be explored further when the two variables are interacted in the regression analysis) and capital market failure, respectively. In addition to diversification, industry innovativeness is positively correlated with most industry types, which is reasonable for a period of technology introduction and economic growth.

[71] Morikawa, *Zaibatsu*. Both Mitsui and Ōkura began as merchandisers (dry goods, groceries), Mitsubishi was a shipping company, Sumitomo and Furukawa were mining companies, and Yasuda started in finance.

[72] The positive correlation with innovative industries may obtain due to the majority of new industries being innovative. See Table 2.

Regression Results

The results from the probit regressions in Table 4 fail to reject the hypothesis that *zaibatsu* affiliation increases the likelihood of first entry into a new industry.[73] This is indicated by the positive coefficient on *zaibatsu* affiliation in Column 1. Including control variables at the firm level (Column 2) and industry level (Columns 3 and 4) reinforces this positive relationship between *zaibatsu* and the probability of first entry. In addition, public listing decreases the likelihood of first entry while diversification (when interacted with *zaibatsu*) increases it, confirming the earlier correlation analysis.

These results are qualitatively similar to those that include population density (Column 5), while factor intensity (Column 6) does not appear to affect the probability of first entry. Here, the positive correlation between *zaibatsu* and first entry is no longer significant, perhaps a consequence of the small sample size.

One minor qualification to *zaibatsu* leadership in pioneering industries is that at the same time *zaibatsu* were less innovative than independent firms. This is indicated by the negative coefficient on the interaction between *zaibatsu* affiliation and industry innovativeness (Columns 3 and 4). In other words, a pioneer is not necessarily an innovator. One interpretation of this result is that *zaibatsu* were more likely to pioneer industries that used technology with demonstrated market viability.[74] Another explanation may lie in the relatively few capital-intensive industries developed in this period (in other words, sectors better able to exploit advantages of *zaibatsu* affiliation). *Zaibatsu* may also have been reluctant to make risky investments due to a lack of entrepreneurial drive in latter generations of owners or to diminishing returns to diversification (in other words, administrative complexity). These possibilities are further discussed in later sections.

Overall, the results suggest that in the early stage of Japanese industrialization, conglomerate membership offered a substantial advantage in pioneering new sectors, even apart from that associated with privately-held ownership. This advantage may issue from their diversified nature and is amplified in less densely populated areas where the risk of new ventures is greater. In addition, I find signs that *zaibatsu* were reluctant to pioneer industries with radically new technologies, which I attribute to their service sector orientation and their entry into industries with greater market power potential.

Robustness

Throughout my analysis, I control for data heteroskedasticity by estimating and reporting Eicker-White standard errors.[75] Furthermore, I perform joint tests of significance to check whether the coefficients on the independent variables are significantly different from zero. All specifications pass at the 1 percent level of significance.

I test for functional form and omitted variable bias with a specification link test. This takes the fitted

[73] An alternative to the probit model, which uses a standard normal distribution to estimate probabilities, is the logit model, which uses a logistic distribution. The results from logit regressions are qualitatively similar to those of the reported probit results, which suggests that results are robust across functional specifications.

[74] While the summary statistics appear to contradict this result, they do not take into account the influences of other variables in the sample.

[75] As mentioned earlier, I cluster the standard errors in the probit regressions using the four-digit JSIC industry codes to account for correlation within the error terms.

values of the residual from the original regression, squares them, and reinserts them into the model as an additional variable. The modified model is estimated to check for significance in the new variable. The null is that the model has no omitted variables, and if correctly specified, the squares of the residuals should not be significant (since they would not show a pattern that could be explained with additional control variables). A significance level above 5 percent is generally interpreted as failure to reject the hypothesis (in other words, model is not incorrectly specified). Aside from the first specification (Column 1), all other specifications are above this threshold, which means that the null hypothesis of no omitted variables cannot be rejected.

Restricted Datasets

I restrict the dataset two ways, first by including only industries that were contested by both *zaibatsu* and independent firms, and second by running separate regressions for firm entry before and after the year 1893, when joint stock firms became legal commercial entities.[76] Both sets of results are shown in Table 5.

By limiting analysis to industries in which both independent firms and *zaibatsu* enter, I can check whether earlier results are due to different industry preferences (for example, *zaibatsu* preferring monopolies or large-scale industries), possible entry deterrence, and the inclusion of monopolies. There are some differences between the full and restricted datasets. Using shared entry with the specification from Column 4 in Table 4, I find that there is a slight positive correlation between *zaibatsu* and first entry, especially when the former is interacted with the diversification variable. This differs from the earlier result where either *zaibatsu* affiliation or its interaction with diversification has a positive effect on first entry, but not simultaneously. In other words, *zaibatsu* affiliation itself, apart from the risk-sharing benefit of diversification, increases chances of leading entry. This may indicate a competitive advantage in mobilizing resources to establish a startup ahead of independent rivals interested in the same industry, and may corresponds to different borrowing costs.

The comparison of results from before and after the year 1893 is to check whether behavior changed after the Japanese government implemented a commercial code that gave legal standing to publicly-listed firms. With more robust institutions delineating fiduciary responsibilities, firms should have found it easier to finance investments externally (for example, equities, loans), blunting the advantage of internal financing in *zaibatsu*. The probit results for the pre-1893 sample show a positive correlation between *zaibatsu* affiliation and first entry overall, and a negative correlation between affiliation and first entry into innovative new industries, similar to results from the full dataset. However, the results from the post-1893 sample show that affiliation is no longer significantly correlated with first entry (although the coefficient remains positive). The negative coefficient on the interaction between *zaibatsu* affiliation and first entry into innovative sectors remains, although not statistically significant. Public listing continues to negatively influence first entry, suggesting the importance of investment autonomy in financing risky ventures despite stronger legal

[76] Notwithstanding the many joint stock companies that existed before the 1890s, prior to the implementation of this code they had no legal basis. A legal commercial code based on German practices was officially adopted in 1891, although only implemented in July 1893. The code established three types of commercial entities, unlimited partnerships, limited partnerships, and joint stock companies. This study makes a distinction between privately-held firms (partnerships) and public firms (joint stock).

provisions for equity finance.[77]

Second Entry

As earlier results have shown, *zaibatsu* appear to lag their independent competitors in leading entry into innovative new industries. This suggests that perhaps *zaibatsu* prefer letting independent firms take risks by entering first and waiting until the new industry is accepted by the market. That is, compared to independent firms, *zaibatsu* may prefer second entry into a new industry. To test this possibility, I rerun all the earlier probit specifications with second entry as the dichotomous dependent variable; the results are shown in Table 6.

Results from some specifications show no significant positive correlation between *zaibatsu* and second entry, implying that *zaibatsu* were not waiting to enter industries pioneered by other firms. This may also suggest stratification of the Japanese market between large conglomerates leading (and dominating) certain industries and smaller firms in others, which was the case in the postwar period. Public listing is again a disadvantage for first entry, but becomes an advantage when interacted with *zaibatsu* (Column 4). This may indicate investor confidence in the activities of these conglomerates.

Discussion and Extensions

One objection to the findings of this study is that small firms may be under-represented in the corporate genealogies, which may bias the results. This may be true especially if they were early entrants in industries but had failed to survive, grow, or be acquired. While the possibility of small firm censorship may exist, in general, I believe that such objections to the present findings are not persuasive. The corporate genealogies include not only direct ancestors of successful contemporary firms, but also unrelated firms whose assets were purchased or absorbed by direct ancestors. That is, the genealogies include asset activity, such as those transferred when a small, possibly innovative firm disbands or becomes bankrupt. This argument is bolstered by the observation that many new industries of the Meiji Period were manufacturing oriented, which typically requires fixed capital. Furthermore, I exclude certain service sectors that are likely to contain establishments undocumented in the genealogies.

Another concern is that while *zaibatsu* may have been more likely to lead entry, their absolute number of first entrants is small, bringing into question their overall impact. This observation assumes erroneously that *zaibatsu* and independent firms were similar, when in reality a single *zaibatsu* affiliate was usually much larger and more productive than an independent firm. As shown in Table 1, many industries (manufacturing and commercial) that *zaibatsu* pioneered were capital-intensive or large scale, and thus out of reach for most independent investors. These industries were also important for production in other sectors (for example, metal mining for machine manufacture), suggesting an alternative means for *zaibatsu* to lead industrialization.

[77] The results from all three restricted samples are also tested for functional form and omitted variable bias, and are above the relevant thresholds for significance. Like the probits for the original sample, standard errors are adjusted for heteroskedasticity and clustered by industry.

Detailed performance measures at the firm level, such as numbers for capitalization, workers, and revenues, would improve this study's findings, but lack of documentation prevents their inclusion. That said, given the relatively meager number of firms and industries used in earlier studies for this time period, one of the strengths of the current dataset is its size, reducing small sample bias. Moreover, the object of this study is to compare corporate behavior via qualitative measures (in other words, entry timing, industry establishment), for which the current dataset is adequate.

Future research

While the main finding of this chapter supports the view that *zaibatsu* assisted the development of industry, it also disputes the notion that they were vanguards of innovation. What accounts for this seemingly important omission in earlier research?

One possible explanation is the emphasis on firm characteristics as opposed to industry-level determinants of performance, and the ease in drawing contrasts between specific conglomerates and independent firms.[78] This paper itself leads with stylized facts about *zaibatsu*'s preponderant size and better access to resources that were deemed critical to pushing forward innovation and expansion. But even these identifying features are misleading. For example, as mentioned in the introduction, *zaibatsu* were less able to attain economies of scope since their holdings were so diverse. This applied especially to the original trading and shipping companies that specialized in services.

Thus, a useful extension to this research would be to compare the role of firm networks and upstream-downstream linkages in the process of Japanese industrialization. It is argued that in postwar Japan, the inability of independent firms to internalize transaction costs with suppliers through acquisition (because of an immature investment environment) motivated them to be efficient.[79] This could be done through coordination with other independent companies to provide goods and services, creating interfirm dependencies and alliances that together reduced any inherent size disadvantages.[80] An expanded dataset that tracked the longevity of both startups and firms established with existing assets could allow one to see whether this scenario applied to prewar industrialization as well.

This paper also remains silent on an important actor: the government. As mentioned earlier, among other policies the government sponsored model factories at the beginning of the Meiji Period to jumpstart industrialization. Later, it privatized these enterprises (with many purchased by *zaibatsu*) and indirectly supported industry through subsidies and contracts. Consequently, assessing the government's contribution to Japanese economic growth is complicated in part to the difficulty of disentangling direct and indirect public support. However, it may be possible to use the current dataset to explore the evolution of the

[78] The idiosyncrasies of individual *zaibatsu*, with their different diversification strategies and founders' colorful personalities, may also have defied collective comparison. There was Mitsubishi's Yatarō transporting government soldiers on their punitive expedition to Taiwan; financial wunderkind Yasuda of the eponymous *zaibatsu* transforming a small money-changing shop into a banking empire; mulishly-focused Furukawa, the mining magnate who excused the purchase of an unprofitable mine as "throwing away 300,000 yen on [his] hobby;" Morikawa, *Zaibatsu*, p. 74.

[79] Gao, *Japan's*.

[80] In a sense, this type of firm coordination would be akin to the strategy behind public-private partnerships in late development theory, writ small instead of at the macro-level.

different sectors started by either the government of the private sector, looking at characteristics like rate of entry or capital intensity. In turn, one may get a better sense of the efficacy of Meiji Japanese industrial policies and how well the government prepared the foundation for later development.

Finally, an understanding of the Japanese precedent is especially valuable for modern China scholars. The Meiji Period saw the unbridled proliferation of a free market system, extensive international trade, and massive transfer of modern technology, all present in present day China. How the Japanese government successfully freed itself of an antiquated economic system and ill-functioning public enterprises and spread its growing wealth throughout the interior of the country are lessons that can be well learned by any developing country, but more exigently by its lumbering East Asian neighbor. Moreover, while it may seem that discussions of industry pioneering are quaint to a global economy where multinationals leave intercontinental imprints and are responsible for introducing new technology, it is reasonable to expect domestic competition in older and new industries, in both manufacturing and services.

Conclusion

It has long been accepted that industrialization in late nineteenth century Japan owed much to the leadership of *zaibatsu*. Using a new dataset of firm establishment dates taken from corporate genealogies, I find evidence that *zaibatsu* are indeed more likely to pioneer new industries, particularly capital-intensive ones. This advantage was likely due to their size and diversified nature, which allow internal financing and reduced investment risk. I also find that public listing decreases the probability of first entry, regardless of conglomerate membership. This result is consistent with such firms' lower autonomy in making investment decisions. Nevertheless, even with these characteristics, *zaibatsu* exhibit technological conservatism in that they were less likely to lead entry into innovative (as opposed to imitative) new industries.

These results highlight an important limitation of earlier analyses of Japanese development, which focus on the development of individual sectors without comparing differences among them (for example, relative risk) or on characteristics of individual firms. A better understanding of industry formation in emerging markets like early modern Japan may require analysis of both intra- and interfirm coordination. Data disaggregated below the national level can also reveal variations that may help policymakers tailor strategies for economic growth.

Appendix: List of New Industries in Meiji Period

Industry Name	JSIC4	Year
Coal mining	611*	1868
Cotton/spun rayon fabric weaving	1441*	1868
Beer	1322	1869
Foreign exchange bank	6124*	1869
Joint stock life insurance company	6711*	1869
Machine reeled raw silk	1421*	1870
Glass containers	2514*	1871
Valves and fittings	2992	1871
Fabricated pipe and fittings	2993	1871
Ball and roller bearings	2994	1871
Piston rings	2995*	1871
Mold and dies, parts and accessories	2996	1871
Coastwise freight transport	4323*	1871
National railway	4011*	1872
Postal services	4711*	1872
Water supply installation, draining work	1133*	1873
Cement	2521*	1873
Ocean transport	4311	1873
Ordinary bank	6121	1873
Fired bricks	2551*	1875
Foreign style paper	1821	1876
Food processing machinery	2961*	1876
Technical colleges	9143*	1876
General civil engineering, building works	911*	1877
Miscellaneous silk reeling plant	1419	1877
Construction and mining machinery/equipment	2931*	1877
Spinning machinery	2951*	1877
Wood working machinery	2962	1877
Wool spinning mill	1423*	1878
Rice cleaning	1261	1879
Wheat flour	1263*	1879
Nitric, phosporitic fertilizers	2011*	1879
Credit cooperative association	6313*	1879
Joint stock fire and marine insurance	6721*	1879
Flat glass	2511	1880

Industry Name	JSIC4	Year
Water supply	3911*	1880
Overseas loans and investment institution	6142*	1880
Credit association and related federation	6313	1880
Air transport, scheduled	4411*	1881
Merchandise forwarding	4621*	1881
Bank for commercial/industrial cooperative	6314	1881
Paperboard	1822	1882
Central bank	6111*	1882
Small business finance corporation	6315	1882
Securities exchange	6631	1882
Medical product preparations	2062*	1883
Canned seafood, seaweed	1221*	1884
Manufactured ice	1341*	1884
Hemp spinning mill	1425	1884
Copper smelting and refining, primary	2711*	1884
Power station	3611*	1884
Soda bicarbonate	2021*	1885
Dairy products	1212*	1886
Miscellaneous paper products	1849	1886
Matches	3486*	1886
Miscellaneous electricity establishment	3619	1886
Lead and zinc metal mining	522*	1887
Steel pipes and tubes	2644	1887
Lead pencils	3443*	1887
Postal order and transfer savings institution	6141	1887
Crude petroleum extraction	711*	1888
Twisting yarns	1431*	1888
Paints	2054*	1888
Watches, clocks and parts; not watchcases	3271*	1888
Miscellaneous musical instruments	3429*	1888
Local railway	4021*	1888
Light vehicle passenger transport	4141*	1888
Miscellaneous business services	8599*	1888
Silk spinning mill	1424	1889
Building brick	2532*	1889

Industry Name	JSIC4	Year
Public financial institution for agriculture, forestry, and fishing	6241*	1889
Construction and mining machinery repair	8213*	1889
Unrefined sugar	1251	1890
Soft drinks, carbonated water	1311*	1890
Transport agency	4631*	1890
Textile sanitary fabric	1498*	1892
Machine dyed/finished silk, rayon fabrics	1462	1893
Mutual life insurance company	6712	1894
Plastics	2037*	1895
Asbestos mining	892*	1896
Sugar refining	1252	1896
Wool fabric weaving	1443	1896
Printing ink	2055	1896
Railroad cars	3121	1896
Long term credit bank	6123	1896
Development financial institution	6143	1896
Agricultural cooperative	6231*	1896
Canvas products	1593*	1897
Printing, not mimeograph	1931	1897
Synthetic dyes, organic pigments	2036	1897
Petroleum refining	2111*	1897
Wire drawing	2648	1897
Coke	2131*	1898
Aluminum smelting and refining, primary	2716	1899
Trust bank	6122*	1899
Basic petrol chemicals	2031	1900
Taxicab operators	4112	1900
Telephone and telegraph, not broadcast	4721*	1901
Miscellaneous industrial organic chemicals	2039	1902
Motor vehicle bodies and trailers	3112	1896
Bicycles and parts	3131*	1903
Metallic springs	2892*	1904
Power and distribution transformer	3012	1904
Miscellaneous stone, sand, gravel quarrying	819*	1905
Soy sauce	1242*	1905

Industry Name	JSIC4	Year
Compound chemical fertilizers	2012	1905
Salt	2025	1905
Nails	2871*	1905
Miscellaneous metal smelting and refining, secondary	2729*	1906
Electric bulbs	3031*	1906
Business consultants	8691	1906
Sulphur mining	831*	1907
Distilled alcohol	1324	1907
Basic livestock feed	1352*	1907
Leather tanning, finishing	2411*	1907
Looms, knitting machinery	2952	1907
Dyeing and finishing machinery	2953	1907
Generators, motors, rotating electrical machinery	3011	1907
Tramway	4022	1907
Refrigerated warehousing	4521*	1907
Fire and marine reinsurance company	6724	1907
Veterinary services	8641*	1907
General sawing and planning mill	1611	1908
Rayon, acetate fiber	2041	1908
Coating metal products	2862*	1908
Medical material preparations	2061	1910
Call loan and bill brokerage, not securities	6411	1909
Pastries, cakes	1272	1910
Miscellaneous fabric weaving	1449	1910
Compressed, liquefied gases	2024	1910
Soaps and synthetic detergents	2052	1910
Aircraft	3151*	1910
Miscellaneous chemical fertilizer mining	839	1911
Organic fertilizer	1353	1912
Miscellaneous industrial inorganic chemicals	2029	1912
Fatty acids, hydrogenated oils, glycerin	2051	1912
Ferro-alloys	2623	1912

*: Innovative Industry (in other words, first four-digit industry established in three-digit industry group)

References

[1] Asia Research, *An Outline of Japanese Industry*. Tokyo, Japan: Ajia Kyokai, 1955.

[2] Audretsch, David, "Asymmetric information, agency costs and innovative entry." CEPR Working Paper 985, London, United Kingdom, 1994.

[3] Bryman, Alan, "Animating the pioneer versus late entrant debate: an historical study." *Journal of Management Studies* 34, no. 3 (1997): 415-438.

[4] Denis, David, Diane Denis, and Atulya Sarin, "Agency theory and the influence of equity ownership structure on corporate diversification strategies." *Strategic Management Journal* 20, (1999): 1071-1076.

[5] Dodwell Marketing Consultants, *Industrial Groupings in Japan*. Tokyo, Japan: Dodwell Marketing Consultants, 1975.

[6] Frankl, Jennifer, "An analysis of Japanese corporate structure, 1915-1937." This Journal 59, no. 4 (1999): 997-1015.

[7] Fruin, Mark, *The Japanese Enterprise System*. Oxford, United Kingdom: Clarendon Press, 1992.

[8] Fuentelsaz, Lucio, Jaime Gomez, and Yolanda Polo, "Followers' entry timing: evidence from the Spanish banking sector after deregulation." *Strategic Management Journal* 23, (2002): 245-264.

[9] Gao, Bai, *Japan's Economic Dilemma: the Institutional Origins of Prosperity and Stagnation*. Cambridge, United Kingdom: Cambridge University Press, 2001.

[10] Gerschenkron, Alexander, *Economic Backwardness in Historical Perspective: A Book of Essays*. Cambridge, Massachusetts: Harvard University Press, 1962.

[11] Gertner, Robert, David Scharfstein, and Jeremy Stein, "Internal versus external capital markets." *Quarterly Journal of Economics* 109, no. 4 (1994): 1211-1230.

[12] Gromb, Denis and David Scharfstein, "Entrepreneurship in equilibrium." CEPR Working Paper 3652, London, United Kingdom, 2001.

[13] Han, Jin, Namwoo Kim, and Hong-Bumm Kim, "Entry barriers: a dull, one-, or two-edged sword for incumbents? Unraveling the paradox from a contingency perspective." *Journal of Marketing* 65, (2001): 1-14.

[14] Hirschmeier, Johannes and Tsunehiko Yui, *The Development of Japanese Business*. Cambridge, Massachusetts: Harvard University Press, 1975.

[15] Jones, HJ, *Live Machines, Hired Foreigners and Meiji Japan*. Vancouver, Canada: University of British Columbia Press, 1980.

[16] Kiyokawa, Yukihiko, "Transplantation of the European factory system and adaptations in Japan: the experience of the Tomioka Model Filature." *Hitotsubashi Journal of Economics* 28, (1987): 27-39.

[17] Lieberman, Marvin and David Montgomery, "First mover advantages." Stanford Business School Research Paper 969, Palo Alto, California, October 1987.

[18] Lilien, Gary and EunsangYoon, "Timing of competitive market entry: an exploratory study of new industrial products." *Management Science* 36, no. 5 (1990): 568-585.

[19] Lockwood, William, *The Economic Development of Japan, Growth and Structural Change 1868-1938*. Princeton, New Jersey: Princeton University Press, 1954.

[20] Loenholm, LH, ed., *Commercial Code of Japan*. Tokyo, Japan: Kokubunsha, 1906.

[21] Long, John, David Rain, and Michael Ratcliffe, "Population density versus urban population: comparative GIS studies in China, India, and the United States." Research Paper, presented at the IUSSP Conference in Salvador, Brazil, August 2001.

[22] Mitchell, Will, "Whether and when? Probability and timing of incumbents' entry into emerging industrial subfields." *Administrative Science Quarterly* 34, (1989): 208-230.

[23] Mitsubishi Public Affairs Committee, *A Brief History of Mitsubishi*. Tokyo, Japan: Mitsubishi Public Affairs, 1990.

[24] Miyajima, Hideaki, Yusuke Omi, and Nao Saito, "Corporate governance and performance in twentieth-century Japan." *Business History Conference On-Line*, `http://www.thebhc.org/publications/ BEHonline/2003/Miyajimaomisaito.pdf`, 2003.

[25] Morikawa, Hidemasa, *Zaibatsu, The Rise and Fall of Family Enterprise Groups in Japan*. Tokyo, Japan: University of Tokyo Press, 1992.

[26] Nakagawa, Keiichiro, "Business strategy and industrial structure in pre-World-War-II Japan." In *The International Conference on Business History: Strategy and Structure of Big Business*. Vol. 1. Edited by Keiichiro Nakagawa. Shizuoka, Japan: Fuji Education Center, 1974.

[27] Ohkawa, Kazushi, *Nihon Choki Tokei Soran* [Long-Term Historical Statistics of Japan]. 5 vols. Tokyo, Japan: Japan Statistical Association, 1987.

[28] Ohkawa, Kazushi and Henry Rosovsky, *Japanese Economic Growth, Trend Acceleration in the Twentieth Century*. Palo Alto, California: Stanford University Press, 1974.

[29] Ohsono, Tomokazu, *Charting Japanese Industry*. London, United Kingdom: Cassell, 1995.

[30] —, "The role of holding companies in prewar Japanese economic development: rethinking zaibatsu in perspectives of corporate government." *Social Science Japan Journal* 4, (2001): 243-268.

[31] Rosovsky, Henry, "Japanese capital formation: the role of the public sector." This Journal 19, no. 3 (1959): 350-375.

[32] Rosovsky, Henry, *Capital Formation in Japan*. New York, New York: The Free Press, 1961.

[33] Russell, Oland, *The House of Mitsui*. Boston, Massachusetts: Little, Brown and Company, 1939.

[34] Schoenecker, Timothy and Arnold Cooper, "The role of firm sesources and organizational attributes in determining entry timing: a cross-industry study." *Strategic Management Journal* 19, (1998): 1127-1143.

[35] Shamsie, Jamal, Corey Phelps, and Jerome Kuperman, "Better late than never: a study of late entrants in household electrical equipment." *Strategic Management Journal* 25, (2004): 69-84.

[36] Smith, Thomas, *Political Change and Industrial Development in Japan: Government Enterprise, 1868-1880*. Palo Alto, California: Stanford University Press, 1974.

[37] Statistics Bureau of Japan, *Standard Industrial Classification of Japan*. Tokyo, Japan: Statistics Bureau of Japan, 1984.

[38] —, "Land and climate." Chap. 1. *Historical Statistics of Japan*. `http://www.stat.go.jp/data/chouki/zuhyou/01-06.xls`, accessed 15 June 2007.

[39] Teranishi, Juro, "The main bank system." In *The Japanese Economic System and its Historical Origins*. Edited by Tetsuji Okazaki and Masahiro Okuno-Fujiwara. Oxford, United Kingdom: Oxford University Press, 1999.

[40] Teranishi, Juro, "Were banks really at the center of the prewar Japanese financial system?" *Monetary and Economic Studies*, March (2007): 49-76.

[41] Travis, Joseph, *The Shipping Industry of Japan*. Washington, DC: Foreign Economic Assocation, 1945.

[42] Trewartha, Glenn, *Japan: A Physical, Cultural, and Regional Geography*. Madison, Wisconsin: University of Wisconsin Press, 1945.

[43] U.S. Census Bureau, *Manufactures*, 1905 edition. Washington, DC: Government Printing Office, 1907.

[44] Yagura, Shintaro and Yoshiro Ikushima, *Shuyo Kigyo no Keifuzu* [Genealogies of Major Corporations]. Tokyo, Japan: Yushodo Publishing, 1986.

[45] Yamamura, Kozo, "Success illgotten? The role of Meiji militarism in Japan's technological progress." This Journal 37, no. 1 (1977): 113-135.

[46] Yushodo Publishing, *Eigyo Hokokusho Shusei* [Business Reports Collection]. First series. Tokyo, Japan: Yushodo Film Publication, 1966.

Table 1: List of Industries with *Zaibatsu* Entry

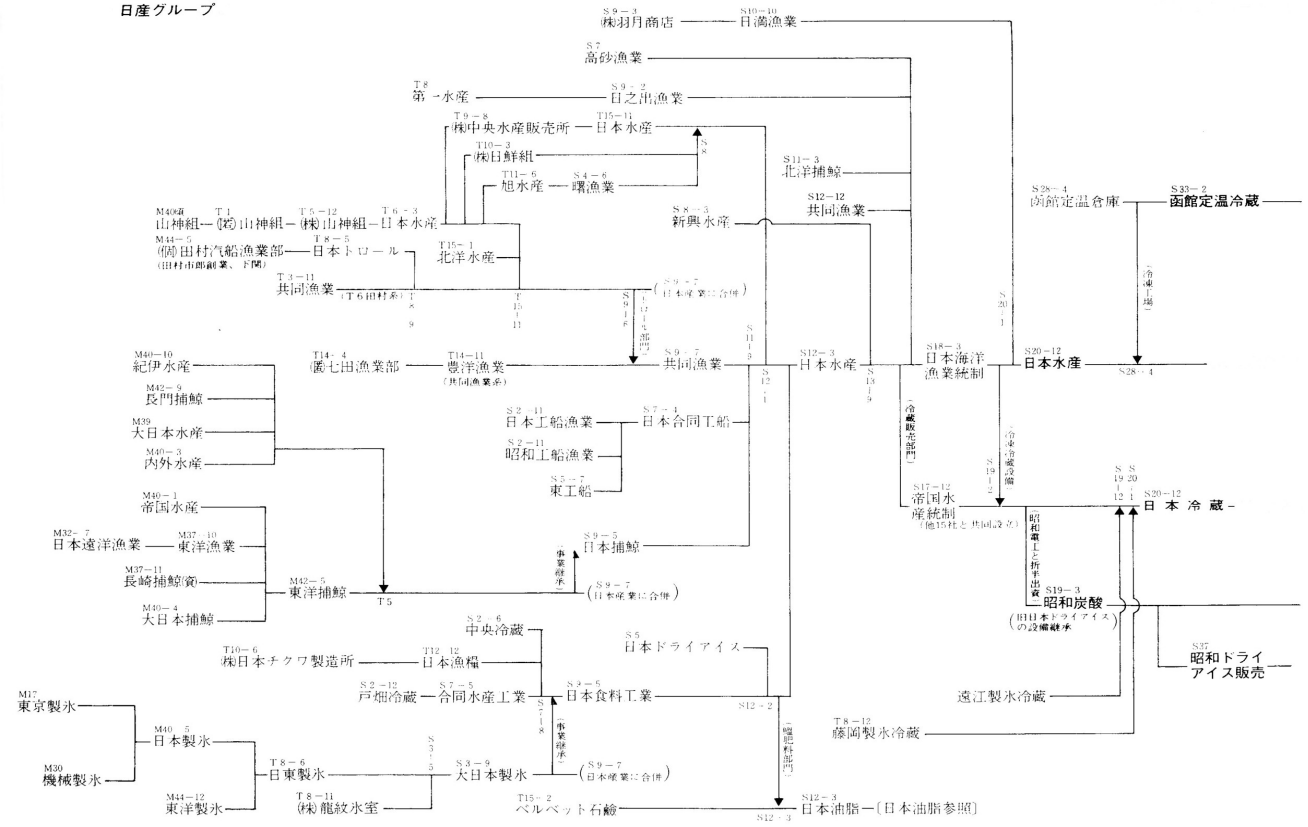
Industry Name	JSIC4	Year	*Zaibatsu*
Coastwise freight transport	4323[a]	1871	Mitsubishi[b]
Ocean transport	4311	1873	Mitsui, Ōkura[b]
Ordinary bank	6121	1873	Mitsui[b]. Sumitomo, Yasuda
Foreign exchange bank	6124	1874	Mitsubishi, Mitsui
Postal services	4711	1875	Mitsubishi
Technical college	9143[a]	1876	Mitsubishi[b]
Joint stock fire and marine insurance company	6721[a]	1879	Mitsui[b], Yasuda
Water supply	3911[a]	1880	Mitsubishi[b]
Credit association and related federation	6312	1880	Yasuda[b]
Life insurance stock company	6711	1881	Mitsubishi
Copper smelting and refining, primary	2711[a]	1884	Furukawa[b], Sumitomo
Lead and zinc metal mining	522[a]	1887	Mitsui[b]
General civil engineering, building works	911	1887	Ōkura
Coal mining	611	1889	Mitsubishi, Mitsui, Sumitomo
Construction and mining machinery repair	8213[a]	1889	Sumitomo[b]
Coke manufacture	2131[a]	1898	Mitsubishi[b]
Crude petroleum production	711	1905	Ōkura
Compound chemical fertilizers	2012	1905	Yasuda[b]
Petroleum refining	2111	1905	Ōkura
Miscellaneous metal smelting and refining, secondary	2729[a]	1906	Furukawa[b]
Business consulting	8691	1906	Ōkura[b]
Aircraft manufacture	3151[a]	1910	Mitsubishi[b]
Local railway	4021	1912	Mitsui

[a]Denotes an innovative industry (ie, first four-digit industry in three-digit industry group.)

[b]Indicates if firm is the first entrant in four-digit industry.

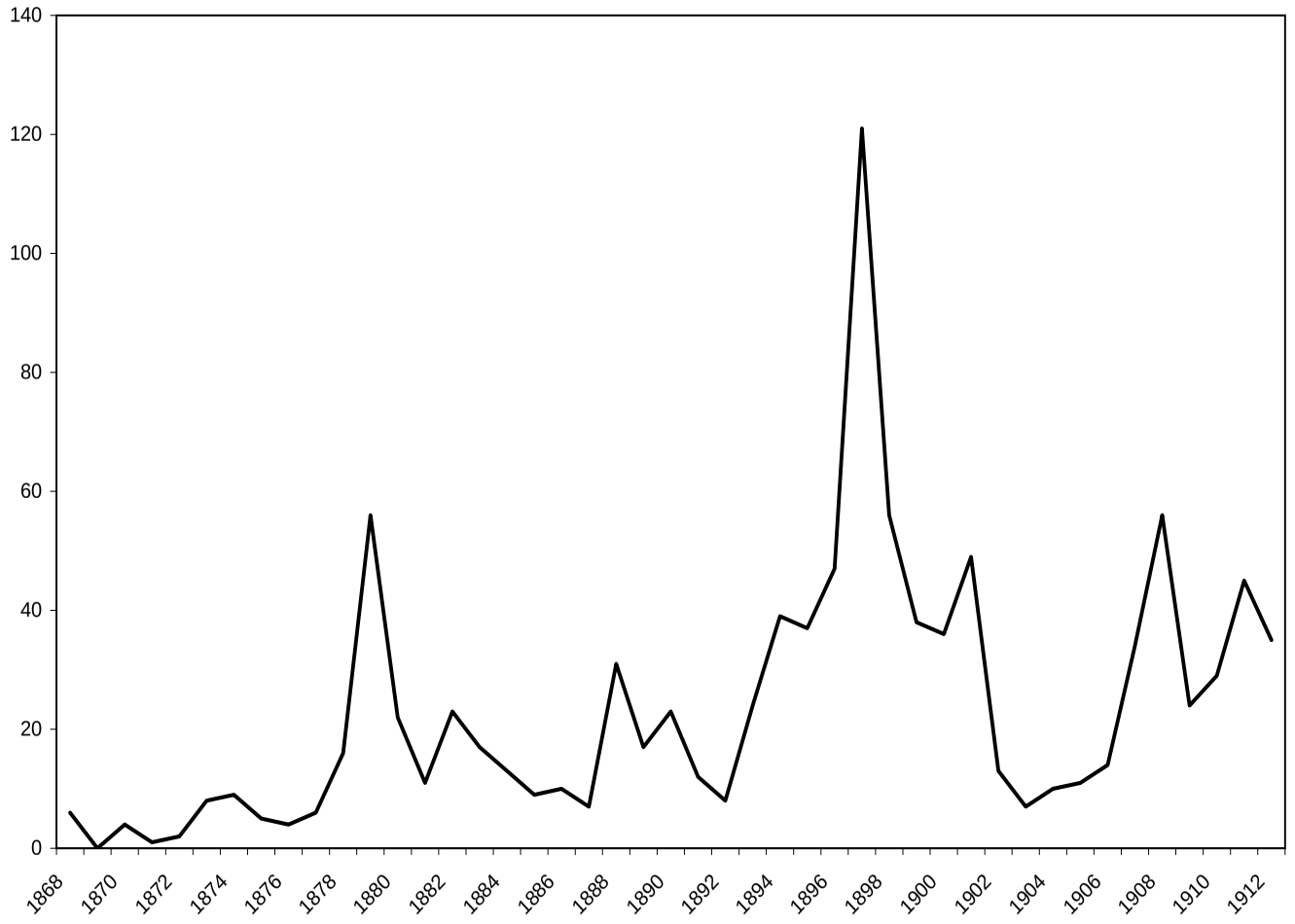
Source: See the text.

Figure 1: Corporate Genealogy Example



Source: Yagura and Ikushima, *Shuyo*.

Figure 2: Number of Startup Establishments, 1868-1912



Source: See the text.

Table 2: Summary Statistics

	Establishment-level (All)[a]			*Industry-level (First Entry)*[b]		
	All	Independent	*Zaibatsu*	All	Independent	*Zaibatsu*
Entry in new 4-digit industries	1,481	1,449	32	111	99	12
Entry in innovative industries[c]	348	322	26	55	47	8
Major industry group						
Agriculture and construction	36	27	9	8	7	1
Manufacturing	222	215	7	71	66	5
Utilities and transportation	152	146	6	15	12	3
Financial services	1,071	1,061	10	17	14	3
Ownership						
Publicly-listed	1,023	1,017	6	51	49	2
Privately-held	170	141	29	42	32	10
Population density[d]						
Urban	425	411	14	69	61	8
Rural	100	100	0	11	11	0

[a]See text for establishment exclusions (eg, government-owned firms, establishments in foreign locations).

[b]The sum of independent and *zaibatsu* industry entries exceeds the total number of industries because of overlaps (ie, industries with shared entry).

[c]In other words, first 4-digit industry to be established within its 3-digit family.

[d]See text for detail about appropriate geographic reference. Urban is a prefecture over 386 people per square kilometer.

Source: See the text.

Table 3: Correlations

	FIRST	ZAIB	DIV	LISTED	INNOV	AGR	MFG	UTIL	FIN	DEN
First entry	1									
Zaibatsu	0.142*	1								
Diversified firm	0.091*	0.832*	1							
Publicly-listed	-0.194*	-0.312*	-0.257*	1						
Innovative industry	0.133*	0.190*	0.197*	-0.159*	1					
Agriculture, construction	0.053*	0.166*	0.207*	-0.092*	0.278*	1				
Manufacturing	0.265*	0.040	0.067*	-0.278*	0.359*	-0.110*	1			
Utilities, transportation	-0.031	0.049	0.057*	0.056*	0.538*	-0.078*	-0.224*	1		
Financial services	-0.228*	-0.132*	-0.177*	0.237*	-0.760*	-0.233*	-0.668*	-0.475*	1	
Population density	0.040	0.024	0.031	-0.023	-0.006	0.061	0.045	-0.076	-0.015	1
Capital-labor expenditure ratio	0.058	0.192*	0.013	-0.011	0.055	-0.034	0.034	–	–	-0.034

*= Significant at the 5 percent level.

Source: See the text.

Table 4: Probit Results

Dependent Variable: First Entry	(1)	(2)	(3)	(4)	(5)	(6)
Zaibatsu	1.170***	0.276	0.502	1.524***	4.388*	3.174
	(0.372)	(0.446)	(0.494)	(0.553)	(2.637)	(2.557)
Diversified		-0.022	-0.019	-0.068*	0.062	-0.084
		(0.043)	(0.043)	(0.041)	(0.135)	(0.173)
Publicly-listed		-0.912***	-0.875***	-0.503***	-0.604***	-0.451*
		(0.135)	(0.158)	(0.178)	(0.230)	(0.240)
Innovative industry			0.392	-0.427	-0.683**	-0.862***
			(0.412)	(0.265)	(0.289)	(0.262)
Major industry groups				included	included	included
Population density					0.034	
					(0.078)	
Capital-labor expenditure ratio						-0.015
						(0.023)
Zaibatsu · Diversified		0.334**	0.319*	0.066	dropped[a]	dropped[a]
		(0.167)	(0.190)	(0.233)		
Zaibatsu · Publicly-listed		0.324	0.291	0.553	2.076	dropped[b]
		(0.861)	(0.875)	(1.123)	(1.605)	
Zaibatsu · Innovative industry			-0.500*	-0.462	0.083	dropped[a]
			(0.486)	(0.477)	(0.984)	
Zaibatsu · Population density					-2.671	
					(1.727)	
Zaibatsu · K/L ratio						-0.070
						(0.074)
N	1,481	1,186	1,186	1,186	381	197
R^2	0.030	0.106	0.121	0.310	0.280	0.125

*= Significant at the 10 percent level.

**= Significant at the 5 percent level.

***= Significant at the 1 percent level.

[a]Variable dropped from specification due to singularity in results (in other words, missing standard error).

[b]Variable dropped from specification due to collinearity with other variable(s).

Notes: Robust standard errors are in parentheses and clustered by four-digit JSIC industry

Source: See the text.

Table 5: Restricted Sample Probit Results

Dependent Variable: First Entry	*Shared*	*Pre*-1893	*Post*-1893
Zaibatsu	1.509*	1.442**	0.743
	(0.917)	(0.725)	(0.586)
Diversified	-0.208*	-0.085	0.035
	(0.111)	(0.061)	(0.071)
Publicly-listed	-0.079	-0.049	-0.508**
	(0.634)	(0.351)	(0.211)
Innovative industry	1.866***	-0.193	-0.707**
	(0.358)	(0.429)	(0.315)
Major industry groups	included	included	included
Zaibatsu · Diversified	0.717***	0.111	dropped[b]
	(0.275)	(0.249)	
Zaibatsu · Publicly-listed	0.469	-0.179	dropped[b]
	(1.019)	(1.048)	
Zaibatsu · Innovative industry	dropped[a]	-0.180	-0.958
		(0.675)	(0.884)
N	786	145	1,041
R^2	0.465	0.184	0.341

*= Significant at the 10 percent level.

**= Significant at the 5 percent level.

***= Significant at the 1 percent level.

[a]Variable dropped from specification due to singularity in results (in other words, missing standard error).

[b]Variable dropped from specification due to collinearity with other variable(s).

Notes: Robust standard errors are in parentheses and clustered by four-digit JSIC industry

Source: See the text.

Table 6: Second Entry Probit Results

Dependent Variable: Second Entry	(1)	(2)	(3)	(4)	(5)	(6)
Zaibatsu	0.825**	0.252	0.143	0.841	-378.396***	18.317
	(0.352)	(0.587)	(0.595)	(0.556)	(11.845)	(11.813)
Diversified		-0.005	-0.005	0.005	-0.033	-0.072
		(0.052)	(0.053)	(0.054)	(0.254)	(0.548)
Publicly-listed		-0.476***	-0.391**	-0.036	-0.283	-0.055
		(0.167)	(0.196)	(0.206)	(0.240)	(0.251)
Innovative industry			0.644*	0.143	0.331	-0.308
			(0.377)	(0.237)	(0.325)	(0.214)
Major industry groups				included	included	included
Population density					0.236***	
					(0.066)	
Capital-labor expenditure ratio						-0.001
						(0.018)
Zaibatsu · Diversified		-0.025	0.054	-0.138	dropped[a]	-0.072
		(0.144)	(0.182)	(0.178)		(0.548)
Zaibatsu · Publicly-listed		1.140	1.046	1.083**	dropped[a]	dropped[b]
		(0.815)	(0.827)	(0.796)		
Zaibatsu · Innovative industry			-0.164	-0.030	dropped[a]	dropped[b]
			(0.669)	(0.686)		
Zaibatsu · Population density					221.540***	
					(7.453)	
Zaibatsu · K/L ratio						-0.751**
						(0.275)
N	1,435	1,146	1,146	1,146	357	169
R^2	0.014	0.037	0.085	0.195	0.096	0.036

*= Significant at the 10 percent level.

**= Significant at the 5 percent level.

***= Significant at the 1 percent level.

[a]Variable dropped from specification due to singularity in results (in other words, missing standard error).

[b]Variable dropped from specification due to collinearity with other variable(s).

Notes: Robust standard errors are in parentheses and clustered by four-digit JSIC industry

Source: See the text.